

Battery-Gated Measurement Validity in Wireless Sensor Deployments: Fault Characterisation, Recovery, and Downstream Consequences

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Abstract—Long-running wireless sensor deployments degrade as their batteries deplete, and the resulting measurements can remain within physically plausible ranges while being entirely uninformative. We characterise this failure on the Intel Berkeley Research Lab deployment (54 motes, 2.3×10^6 readings) and show that the conventional remedy—filtering readings by plausible range—leaves the corruption intact in the evaluation window. Three results follow. First, the fault is not a smooth voltage-dependent drift but a discrete *rail fault*: the converter saturates to a fixed 122.1°C with humidity simultaneously railing negative, affecting 383,842 readings. Gating on battery voltage ($V \geq 2.4$ V) reduces the temperature standard deviation from 12.42°C to 3.24°C, removing the corruption rather than clipping it. Second, a voltage-free detector that fuses temperature, humidity and light into a single Mahalanobis score independently validates the gate—it identifies only four faults the gate misses in 2.2×10^6 readings—while showing the gate over-discards 8.59% of readings; a four-stage cleaner exploiting this recovers 10.9% more data while *improving* accuracy by 3.6%. Third, the measurement-validity protocol determines the outcome of downstream estimation: under range filtering, graph message passing in a Kalman filter appears 13% *worse* than a graph-free filter at reconstructing unobserved nodes, whereas under voltage gating the same comparison reverses to a $27.1 \pm 2.9\%$ improvement. We further characterise reconstruction of removed measurements, showing global low-rank completion halves the error of graph harmonic extension, and quantify the point at which undetected corruption defeats both. Applying the identical estimator unchanged to human GPS mobility trajectories (taxi and private-car trips, GeoLife dataset) reproduces the graph benefit and surfaces a second, independent mechanism for protocol-dependent conclusions: taxi’s fitted coupling coefficient is the largest observed in either domain, yet is invisible in a held-out test split that happens to contain no co-presence to exercise it.

Index Terms—Measurement validity, sensor fault detection, wireless sensor networks, data quality, Kalman filtering, matrix completion, battery depletion.

I. INTRODUCTION

WIRELESS sensor deployments intended to run unattended for weeks or months are subject to a failure mode that conventional quality control handles poorly. As a node’s battery depletes, its measurements degrade gradually and then catastrophically—but the degraded values need not

leave the physically plausible range, and so survive the range checks typically applied before analysis. The consequence is not merely noisier data. Because depletion correlates with time, and because evaluation splits in time-series work are usually chronological, the corruption concentrates in precisely the portion of the record reserved for evaluation.

Cleaning sensor streams is well studied—ensemble and stream-based models have been proposed for precisely this setting [8], and taxonomies of sensor fault types are established [6], [7]. What has received less attention is the *consequence* of the cleaning choice: whether the protocol adopted before analysis can alter, rather than merely degrade, the conclusion drawn about a downstream method.

This paper characterises that failure on the Intel Berkeley Research Lab deployment, the most widely used public wireless sensor dataset, and traces its consequences through to a downstream estimation task. Our contributions are:

- 1) **Fault characterisation.** We show the corruption is a discrete rail fault rather than a drift, identify its joint-channel signature, and establish battery voltage as the causal index that separates valid from invalid measurements (Section III).
- 2) **Independent validation.** We construct a detector that fuses the three sensing channels without observing voltage, and use it together with a spatial-consensus arbiter to establish that the voltage gate is sound as a detector but over-conservative as an exclusion rule (Section IV).
- 3) **A staged validity cleaner** that separates the two failure modes present in the marginal voltage band, recovering 10.9% more readings while improving downstream accuracy (Section V).
- 4) **Demonstration that the protocol determines the conclusion.** On a graph-based Kalman filtering task, the measured benefit of relational structure reverses sign—from 13% harmful to 27.1% beneficial—according to which validity protocol is applied (Section VI).
- 5) **Characterisation of recovery** for removed measurements, including the regime boundaries between graph-local and global methods, their tolerance to undetected

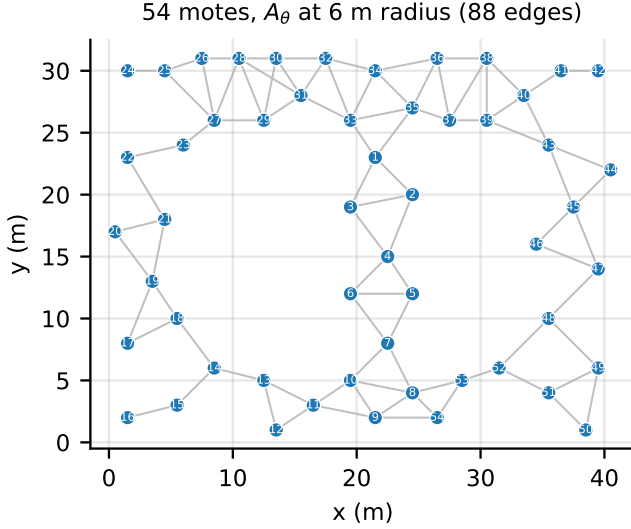


Fig. 1. The 54 motes at their published coordinates, with the adjacency A_6 drawn at a 6 m connection radius. Six metres is the smallest radius yielding a connected graph; at 5 m the network fragments into seven components with two isolated motes.

corruption, and a locality-stratified account of accuracy and precision (Sections VII–IX).

- 6) **External validity check.** We apply the identical graph state-space model, unmodified, to human GPS mobility trajectories (taxi and private-car trips, GeoLife dataset), showing the graph term again beats persistence and finding a second, independent mechanism by which a real, well-fitted effect can be invisible in a held-out test protocol (Section X).

II. DEPLOYMENT AND MEASUREMENT CHAIN

A. Data

The deployment comprises 54 Mica2Dot motes recording temperature, humidity, light and battery voltage over 36 days from 28 February 2004, yielding 2,313,682 readings in the released file. Three mechanical issues require attention before analysis. The `moteid` field contains out-of-range values (6485, 33117, 65407) originating from corrupted packets; 113,438 duplicate (`moteid`, `epoch`) pairs are present; and the `epoch` counter is 16-bit and wraps at 65535, so it cannot serve as a monotonic index across the record. We index by wall-clock timestamp. Sampling is nominally 31 s but irregular in practice (median 35.2 s) and unsynchronised across motes, so readings are resampled to a uniform five-minute grid.

B. Spatial structure

The published mote coordinates permit an adjacency grounded in physical geometry rather than inferred from the signal itself. Median nearest-neighbour spacing is 3.6 m over a 40.5×31 m floor. We threshold Euclidean distance at 6 m—the smallest radius leaving the graph connected—yielding 88 edges and mean degree 3.3 (Fig. 1).

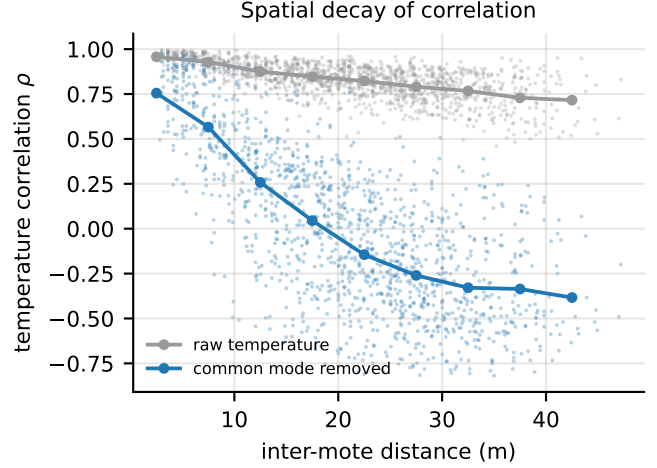


Fig. 2. Temperature correlation against inter-mote distance. Raw correlation (grey) is nearly flat, being dominated by a building-wide diurnal cycle. Removing the network-wide mean (blue) recovers the expected spatial decay. Markers denote bin means.

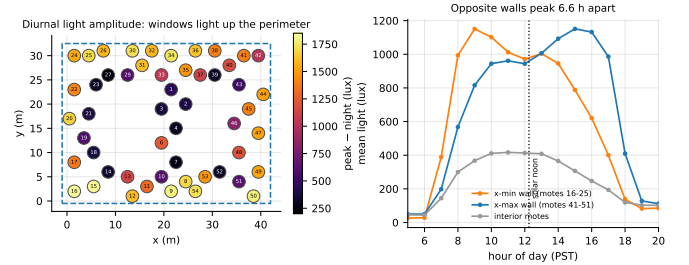


Fig. 3. Left: diurnal light amplitude by mote position; the perimeter is clearly delineated. Right: mean light against hour of day for the two x -extreme walls, which peak 6.6 h apart.

Using geometry rather than signal correlation is deliberate. Fig. 2 shows raw temperature correlation is nearly flat in distance ($\rho = 0.739$ below 6 m against 0.660 beyond 30 m; distance–correlation -0.10), because a building-wide diurnal cycle dominates every mote irrespective of separation. A correlation-thresholded graph overlaps the spatial graph at Jaccard 0.08. Removing the network-wide common mode recovers the expected decay (distance–correlation -0.48). Since inferring the graph from the signal one intends to predict is in any case circular, the published coordinates are the sounder basis.

C. Verification that spatial structure is recoverable

As an independent check that the field carries the spatial structure the adjacency assumes, we recover deployment geometry from the sensing channels alone. Diurnal light amplitude (peak minus night baseline) correlates -0.713 with distance to the nearest wall: perimeter motes average 1569 lux of swing against 511 lux for interior motes (Fig. 3, left). Peak light hour further separates the two x -extreme walls by 6.6 h (08:12 against 14:48), the signature of opposite solar exposure,

TABLE I
CORRUPTION AS A FUNCTION OF BATTERY VOLTAGE. “IMPLAUSIBLE”
DENOTES TEMPERATURE OUTSIDE $[15, 35]^{\circ}\text{C}$, AN INDOOR OFFICE BAND.

Voltage bin (V)	Readings	Implausible (%)
(2.0, 2.2]	232,194	99.96
(2.2, 2.3]	169,129	89.59
(2.3, 2.4]	221,805	20.65
(2.4, 2.5]	398,154	0.00
(2.5, 2.6]	447,827	0.22
(2.6, 2.7]	717,451	0.29

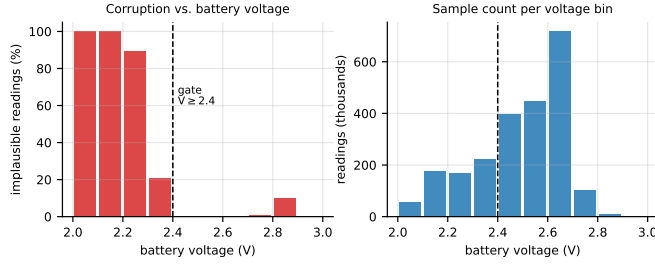


Fig. 4. Left: fraction of implausible temperature readings against battery voltage; the transition approximates a step at 2.4 V. Right: sample counts per bin.

from which the building’s east–west axis can be identified—information absent from the published coordinates, which are given in an arbitrary lab frame. Temperature confirms this independently: diurnal swing correlates -0.519 with distance to wall (7.24°C at the perimeter against 4.80°C inside). Two unrelated channels agreeing supports treating spatial proximity as a meaningful prior.

III. FAULT CHARACTERISATION

A. Battery voltage indexes measurement validity

The temperature channel degrades as batteries deplete, reaching values as high as 385°C . Table I and Fig. 4 show that battery voltage separates valid from invalid readings far more sharply than the readings separate themselves: below 2.3 V almost every reading is implausible, above 2.4 V almost none is.

Gating at $V \geq 2.4$ V retains 72.8% of readings and reduces the implausible fraction to 0.30%. The decisive statistic is dispersion: the temperature standard deviation falls from 12.42°C to 3.24°C . A range filter cannot achieve this, because drifting readings that remain inside $[0, 50]^{\circ}\text{C}$ survive it and silently contaminate the record.

B. The fault is a rail, not a drift

Fig. 5 establishes the mechanism. The corruption is not a smooth voltage-dependent drift, which a calibration curve could invert, but a discrete *rail fault*: the converter saturates to a fixed 122.1°C while humidity simultaneously rails negative. 383,842 readings carry this joint signature. A railed reading retains no recoverable information and cannot be corrected—but it is deterministically detectable, which is what makes the marginal band separable:

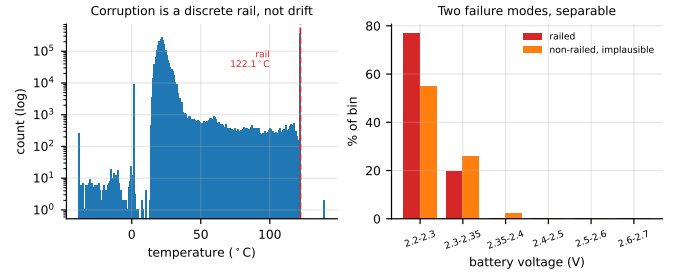


Fig. 5. Left: the temperature histogram is bimodal, exhibiting a discrete spike at 122.1°C rather than a smeared drift—the converter saturates. Right: split by rail status, the $(2.35, 2.40]$ V band is 0.12% railed and its survivors are 2.35% implausible, as clean as the trusted region.

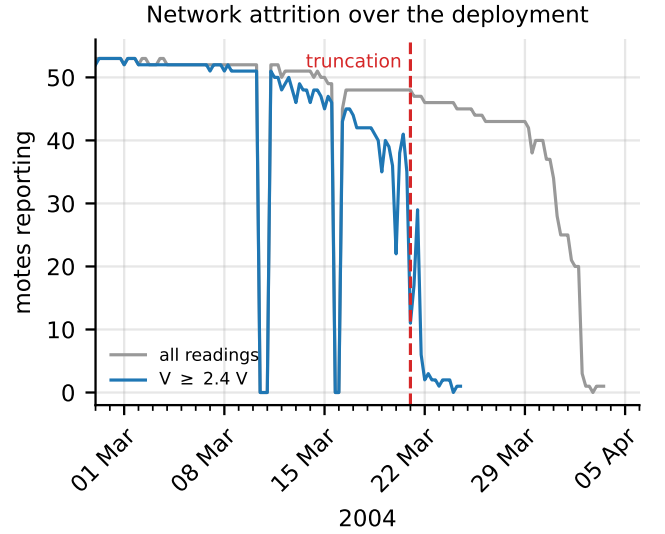


Fig. 6. Motes reporting over time. The deployment decays progressively and is effectively inactive after 23 March. A chronological 70/30 split therefore yields a test set that is only 5.4% observed.

Voltage (V)	n	Railed	Non-railed implausible
(2.30, 2.35]	105,941	19.67%	25.97%
(2.35, 2.40]	115,864	0.12%	2.35%
(2.40, 2.50]	398,154	0.00%	0.00%

The $(2.35, 2.40]$ band is 0.12% railed. It was being discarded for sharing a bin edge with the genuinely broken data below it.

C. Network attrition

A second hazard compounds the first. Motes drop out progressively (Fig. 6) until essentially nothing reports. A chronological 70/30 split over the full record produces a test set with 5.4% fill against 80.9% in training. We truncate to the window in which at least 30% of motes report, giving 6,521 five-minute steps from 28 February to 21 March at 76.6% fill.

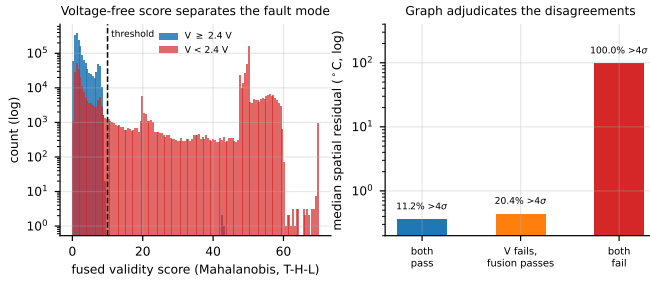


Fig. 7. Left: the voltage-free fused score cleanly separates a healthy mode from a fault mode, with the sub-threshold population concentrated in the latter. Right: spatial residual for each agreement stratum; readings the voltage rule rejects but fusion accepts resemble agreed-good data, not agreed-bad data.

IV. INDEPENDENT VALIDATION BY SENSOR FUSION

The voltage gate was derived from the data it screens and is therefore self-justifying. This section constructs a second detector that never observes voltage, and introduces a third arbiter—the spatial graph—to adjudicate between them. Device-level trust may also be established cryptographically, for instance through hardware-assisted authentication [12]; the approach taken here is complementary, inferring trustworthiness statistically from the measurements themselves, and applies equally to archived data from deployments that carry no such hardware. The adjudication step follows the principle of resolving disagreement by census across independent observers [11].

A. A voltage-free fused validity score

We fuse the three sensing channels into one scalar per reading via the Mahalanobis distance in (T, H, L) , with location, robust scale and correlation estimated on trusted readings:

$$s = \sqrt{z^T C^{-1} z}, \quad z = (x - \mu) \oslash \sigma, \quad (1)$$

with $\mu = [21.93^\circ\text{C}, 39.42\%, 165.6 \text{ lux}]$ and $\sigma = [2.73, 6.03, 240.7]$ (scaled median absolute deviation). The physical anchor is the temperature–humidity correlation of -0.722 : indoor temperature and relative humidity are strongly anti-correlated, so a reading violating that relation is suspect even when each channel is individually within range. This is what allows a single scalar to suffice.

Healthy readings ($V \geq 2.4$) have median score 1.59; sub-threshold readings 49.29; failed readings 50.08. The score attains AUC 0.9971 against the rail signature but only 0.8589 against the voltage rule. That gap is informative: the fused score reproduces the *fault* almost perfectly while only partly reproducing the *rule*.

B. Agreement and graph adjudication

Table II shows the two detectors agree on 91.41% of readings. Critically, the voltage gate misses almost nothing: only four readings in 2.2×10^6 pass the voltage rule while failing fusion. But it rejects 189,843 readings (8.59%) that the sensing channels consider valid.

TABLE II
AGREEMENT BETWEEN THE VOLTAGE RULE AND THE VOLTAGE-FREE FUSED DETECTOR, $n = 2,210,084$ READINGS.

	Fusion pass	Fusion fail
$V \geq 2.4$ (pass)	1,608,565	4
$V < 2.4$ (fail)	189,843	411,672

TABLE III
SPATIAL RESIDUAL AGAINST HEALTHY NEIGHBOURS FOR EACH AGREEMENT STRATUM (NULL STANDARD DEVIATION 0.326°C).

Stratum	n	Median $ r $	Beyond 4σ
Both pass	236,405	0.358°C	11.2%
V fails, fusion passes	5,746	0.436°C	20.4%
Both fail	2,011	97.268°C	100.0%

Table III adjudicates these disagreements using the spatial graph, which is independent of both detectors. The over-discarded readings have median residual 0.436°C , close to agreed-good data (0.358°C) and far from agreed-bad data (97.268°C). They are usable, if modestly noisier (20.4% against 11.2% beyond 4σ).

The conclusion is twofold and is what motivates Section V: the voltage gate is *validated as a detector* and *refuted as an exclusion rule*, by an instrument sharing no information with the rule it evaluates.

C. Two negative results

Two natural hypotheses do not survive testing, and we report them because each would otherwise be an attractive design choice.

Fault simultaneity is not a common-cause event, but only phase conditioning reveals this. Pooled over the record, 35.7% of 30-minute windows exhibit more than eight simultaneous faults, suggesting system-level failure. Conditioned on deployment phase, the picture inverts: before 15 March, 63% of windows are fault-free and 95.5% of those containing a fault involve exactly one mote, with none exceeding eight; after 15 March, only 2% are fault-free and 67.1% involve more than eight, with median faulty fraction 0.93. Early faults are isolated device failures; late co-occurrence is correlated end-of-life as batteries deplete together. The pooled statistic would mislead.

Fusion provides no early warning. Voltage crosses 2.4 V before the sensing channels corrupt in 47 of 51 motes, with median lead 3.0 days. The sensing channels are downstream of the battery, so the fused score confirms the causal mechanism rather than anticipating it. Where early warning is the objective, voltage is already the best available signal.

V. A STAGED VALIDITY CLEANER

Section IV established that the voltage cutoff over-discards. We therefore replace it with a four-stage procedure. The staging follows the ensemble principle applied to sensor-stream cleaning in [8], in which independent weak tests are composed rather than a single decision boundary being tuned; here the

TABLE IV
DETECTIONS BEYOND THE A-PRIORI BOX, ADJUDICATED BY THE GRAPH
(NULL STANDARD DEVIATION 0.326°C).

Rule	Caught	Median $ r $	Beyond 4σ
Empirical box (T, H)	697	3.285°C	79.2%
T - H ellipse	532	8.954°C	87.2%
Box $\cup$ ellipse	900	6.323°C	81.6%
Retained (combined)	241,210	0.359°C	11.2%
Retained (a-priori box)	242,110	0.361°C	11.4%

tests are deliberately heterogeneous—one deterministic, one distributional, one relational, one probabilistic—so that their failure modes do not coincide.

- S1 **Rail detection.** Deterministic joint-channel test: $|T - 122.13| < 0.05$ or $H < 0$. No threshold tuning is required.
- S2 **Validity bounds.** A physical-range guard, tightened using the agreed-healthy population (Section V-A).
- S3 **Graph consensus.** Each surviving candidate is compared against the median of its *trusted* neighbours at the same timestamp. The null is calibrated on trusted data only, so the threshold does not inherit contamination from the candidates it screens; the resulting robust spatial residual standard deviation is 0.518°C, and candidates beyond $\tau\sigma$ are rejected.
- S4 **Heteroscedastic measurement noise.** Surviving marginal readings enter downstream estimation with an inflated noise variance rather than being accepted or rejected outright.

That recovered readings are genuinely sound, rather than merely plausible, is confirmed by their spatial residual: 0.503°C for the [2.35, 2.40] V band against 0.467–0.493°C for trusted readings.

A. Tightened validity bounds

The a-priori guard $T \in [10, 45]^\circ\text{C}$, $H \in [0, 100]\%$ is loose. We calibrate tighter bounds on readings both detectors accept, at 99.9% coverage, which bounds false rejection on good data at 0.1% by construction.

We note that the correlation-aware region is *not* tighter by volume than a percentile box: at matched coverage the empirical box occupies 0.205 of the a-priori volume against 0.923 for the fused ellipsoid, because the light channel is wide-ranging and weakly correlated with the others. Restricting to the (T, H) subspace does not reverse this (box 847.2 against ellipse 890.6). Volume, however, is the wrong figure of merit. Table IV shows both rules detect genuine faults—79–87% of flagged readings lie beyond 4σ against a baseline of 11.4%—and that they catch overlapping but non-identical sets. The ellipse contributes approximately 200 detections no axis-aligned bound can find: the physically impossible hot-and-humid and cold-and-dry combinations. The cost is 0.37%

TABLE V
STAGED CLEANER AGAINST THE HARD VOLTAGE CUTOFF.

	Cutoff	Staged	Change
Rows retained	1,677,047	1,860,173	+10.9%
Grid cells	269,943	294,803	+9.2%
Usable window	5,996	6,390 steps	+33 h
Imputation RMSE	2.147°C	2.070°C	−3.6%

TABLE VI
ONE-STEP-AHEAD (5 MIN) PREDICTION RMSE AT OBSERVED NODES.

Model	RMSE (°C)
Persistence	0.269
KF, no graph ($\theta_{sp} = 0$)	0.275
KF, graph message passing	0.262

of retained data with residual essentially unchanged. The recommended S2 rule is therefore the disjunction

$$T \notin [14.58, 35.91] \vee H \notin [16.08, 55.79] \vee m_{TH} > 4.99. \quad (2)$$

B. Result

Table V shows the cleaner is denser *and* more accurate: whatever additional noise the recovered readings carry, the density gain more than repays it. We note that S4 is inert at the operating point used here ($R_0 = 10^{-4}$ in standardised units, where process noise dominates); it is retained because it becomes load-bearing at larger R .

VI. DOWNSTREAM CONSEQUENCE

We now show that the validity protocol, not the estimator, determines the conclusion one draws about a downstream method.

A. Estimator

We adopt the graph state-space formulation of [1], in which inputs, states and outputs are attributed graphs. For a fixed deployment the time-varying-topology machinery is inactive and the model reduces to

$$s_t = \theta_{tm}s_{t-1} + \theta_{sp}\bar{A}_\theta s_{t-1} + \eta_{t-1}, \quad y_t = s_t + \nu_t, \quad (3)$$

with $\bar{A}_\theta = D^{-1/2}(I + A_\theta)D^{-1/2}$, identity readout, and θ fitted by least squares on the training split. The transition being linear, the filter is an exact Kalman filter. Setting $\theta_{sp} = 0$ yields the graph-free ablation. The error covariance $P_t \in \mathbb{R}^{|V| \times |V|}$ is a covariance *across nodes*: its off-diagonal entries are the only channel by which an observation at one mote informs the estimate at another.

B. The conclusion reverses

Table VII is the central result. Under the range filter, the graph term *degrades* performance by 13%; under the voltage gate, the identical comparison shows a $27.1 \pm 2.9\%$ improvement, favouring the graph in all ten draws (Fig. 8). The conclusion one would draw about the method reverses on a measurement-validity choice.

TABLE VII

IMPUTATION RMSE FOR NEVER-OBSERVED MOTES (9 OF 39 WITHHELD), MEAN OVER 10 RANDOM DRAWS. THE PROTOCOLS DIFFER ONLY IN THE VALIDITY RULE.

Validity protocol	No graph	Graph MP	Change
Range filter $[0, 50]^{\circ}\text{C}$	3.875	4.377	+13.0%
Voltage gate $V \geq 2.4 \text{ V}$	2.519	1.833	−27.1%

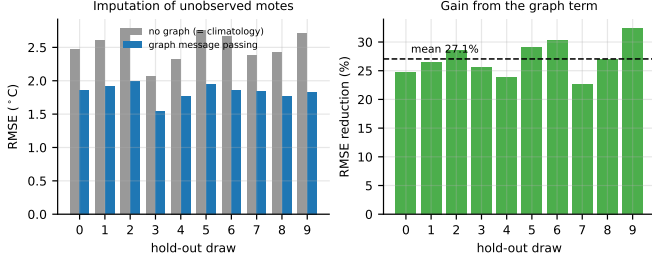


Fig. 8. Left: imputation RMSE per hold-out draw. Right: RMSE reduction attributable to the graph term, mean 27.1%, positive in all ten draws.

The graph-free filter matches climatology to three decimal places, and necessarily so: with no measurement ever reaching a withheld node, its estimate decays to the prior mean. Fig. 9 shows this directly.

Table VI shows the benefit is confined to unobserved nodes; at observed nodes the graph term is worth under 5%. The fitted parameters explain why: $\theta_{tm} = 0.992$ against $\theta_{sp} = 0.0064$. At a five-minute cadence indoor temperature is a near-perfect random walk, so a mote’s own previous reading is very nearly a sufficient statistic.

VII. RECOVERY OF REMOVED MEASUREMENTS

Railed readings are removed, not corrected. We characterise their reconstruction.

A. Graph harmonic extension

Zeroing removed cells and iterating neighbour averaging with observed nodes clamped converges to the harmonic extension—the minimiser of the Dirichlet energy $\sum_{uv} w_{uv}(x_u - x_v)^2$ subject to the observed boundary. Three properties hold empirically. Initialisation is irrelevant: zeros, climatology, random $\mathcal{N}(0, 3)$ and a +100 offset converge to the same fixed point to within 2.8×10^{-17} . The clamp is essential: leaving zeroed cells in as sources degrades RMSE from 0.484 to 1.923°C and collapses the field to standard deviation 0.024°C. Convergence is fast, the maximum per-iteration update falling below 10^{-3} by iteration 8.

Capacity degrades gracefully: at 80% of nodes blank, harmonic extension achieves 1.14°C against 3.32°C for climatology and 2.40°C for single-hop averaging (Fig. 11, left).

B. Global low-rank completion

The field is very low rank: rank 1 explains 81.9% of variance, rank 3 95.2%, with effective rank 1.47 of 41 (Fig. 10,

TABLE VIII

RECONSTRUCTION RMSE UNDER MISSINGNESS SAMPLED FROM THE EMPIRICAL GAP-LENGTH DISTRIBUTION (30%, MEAN OF 3 DRAWS).

Method	RMSE ($^{\circ}\text{C}$)
Climatology	3.137
Harmonic extension	0.926
Low-rank ALS	0.475
Low-rank + graph prior	0.452

TABLE IX

RMSE ($^{\circ}\text{C}$) AT RECONSTRUCTED NODES WITH 30% OF NODES BLANK, AS A FRACTION OF THE *observed* NODES IS SILENTLY RAILED AND TREATED AS TRUSTWORTHY.

Corrupt	Mean agg.	Median agg.	Detect first	Climatology
0%	0.892	0.992	1.064	3.418
5%	8.724	5.123	0.960	3.049
10%	19.067	14.367	1.535	3.390
20%	31.587	29.871	4.516	2.993
50%	59.732	62.791	32.884	3.363

left). Factorising $X \approx UV^T$ and minimising

$$\|K \circ (UV^T + b - Y)\|^2 + \alpha(\|U\|^2 + \|V\|^2) + \lambda_g \text{tr}(V^T L V) \quad (4)$$

gives a 51.2% reduction over harmonic extension (Table VIII). We note this must be optimised by alternating least squares rather than a generic first-order method; Adam produced non-monotone behaviour in the rank ($r=3$: 0.898, $r=8$: 1.407, $r=12$: 0.550), which is optimiser noise rather than a property of the model.

The method has one clean failure mode. For nodes observed at *no* timestamp, plain low-rank collapses to climatology (3.438 against 3.454), as such a node has no data constraining its loading V_j . The graph prior supplies one and recovers much of the loss (1.307), but harmonic extension still wins outright (0.825). The two are complementary, and we route by observability. Reconstruction from an under-determined observation set is ill-posed in the same sense encountered in edge re-identification tasks [10], and is likewise resolved by imposing structure—here a low-rank factorisation and a graph prior—rather than by additional data.

VIII. LIMITS: TOLERANCE TO UNDETECTED CORRUPTION

Table IX quantifies what happens when detection fails. A railed node lies some 30 standard deviations from the field, and the harmonic solution is a weighted average of *all* boundary values, so local damage becomes global. At 5% undetected corruption, reconstruction is already $2.9\times$ worse than predicting the global mean.

Robust aggregation is not an adequate remedy: median aggregation recovers only a factor of approximately 1.4 and is itself worse than climatology from 5% onward. The reason is that the breakdown point is *local*. A median tolerates corruption below half of each node’s own neighbourhood, and this graph is sparse—mean degree 3.3, with 13 of 54 motes unable to outvote even one corrupted neighbour. A 5% global fraction therefore concentrates into a local majority somewhere

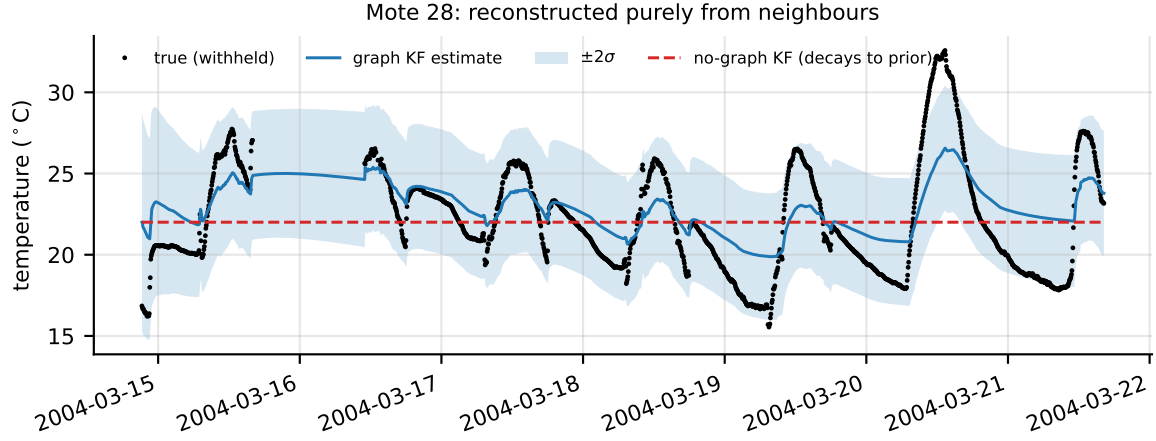


Fig. 9. Mote 28, withheld from the filter for the entire test period and reconstructed from its neighbours through the state-transition graph. The graph-free filter (red, dashed) decays to the prior mean because no measurement can reach the node; the graph filter (blue) tracks the withheld signal, with the shaded band showing $\pm 2\sigma$ from the propagated covariance.

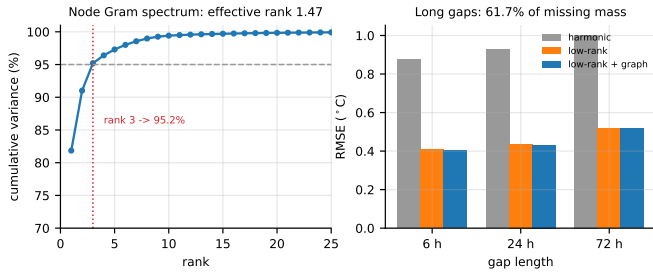


Fig. 10. Left: the node Gram spectrum is dominated by one mode; rank 3 captures 95.2% of variance and the effective rank is 1.47 of 41. Right: on long gaps, which carry 61.7% of all missing cells, low-rank completion approximately halves the error of harmonic extension.

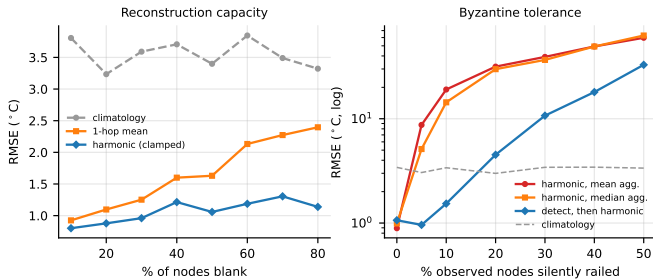


Fig. 11. Left: reconstruction RMSE against the fraction of blank nodes. Right: RMSE at reconstructed nodes as observed nodes are silently failed (log scale). Robust aggregation does not avert the collapse; detection does.

in the graph with near certainty, and the classical $n \geq 3f + 1$ intuition misleads: the relevant quorum is the neighbourhood, not the network.

This is an empirical counterpart to the aggregation lower bounds derived in [9], where the attainable precision of distributed sensor fusion under Byzantine faults is shown to be governed by the structure over which aggregation occurs rather than by the global fault fraction alone. The present

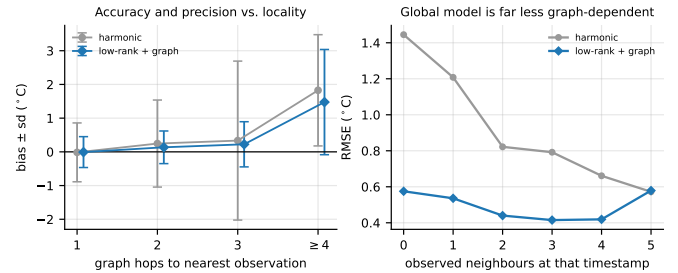


Fig. 12. Left: bias $\pm$ standard deviation against graph-hop distance to the nearest observation. Right: RMSE against the number of observed neighbours.

TABLE X
BIAS / STANDARD DEVIATION / RMSE (°C) BY GRAPH-HOP DISTANCE TO THE NEAREST OBSERVED NODE.

Hops	n	Harmonic	Low-rank + graph
1	58,387	-0.015 / 0.873 / 0.873	-0.007 / 0.458 / 0.458
2	4,384	+0.246 / 1.291 / 1.315	+0.135 / 0.484 / 0.502
3	370	+0.334 / 2.358 / 2.382	+0.224 / 0.670 / 0.706
≥ 4	65	+1.826 / 1.650 / 2.461	+1.476 / 1.561 / 2.149

measurements make the same point concretely for a physical deployment: the governing quantity is the degree distribution, and reporting a network-wide tolerance figure without it would overstate robustness by an order of magnitude.

Screening before reconstruction holds RMSE at 0.96–1.54°C through 10% contamination, an order of magnitude better than robust aggregation. This is the operational justification for the deterministic S1 test.

IX. LOCALITY-STRATIFIED ACCURACY AND PRECISION

Aggregate RMSE obscures how error relates to local graph context. Reporting accuracy (bias) and precision (standard deviation) separately (Table X, Fig. 12) yields three observations.

The graph is a strong hypothesis for the local method and a weak one for the global method: harmonic error grows $2.7\times$ from one hop to three, low-rank only $1.5\times$. A global model reaches an isolated node through the shared factor rather than through the graph.

Accuracy and precision degrade differently, and only bias is graph-directional. Bias is near zero at two to four observed neighbours and turns positive as the neighbourhood empties, isolated reconstructions being pulled toward the field mean. Precision degrades monotonically with isolation for harmonic ($0.516 \rightarrow 1.419$) but is nearly flat for low-rank ($0.414 \rightarrow 0.552$).

Graph position nonetheless predicts an individual cell’s error poorly: correlation between absolute error and hop distance is $+0.089$ for harmonic and $+0.138$ for low-rank. Graph context shifts the error *distribution* substantially while predicting any single cell weakly, so a per-cell confidence interval derived from graph position alone would be poorly calibrated. The ≥ 4 -hop stratum contains only 65 cells and should be read as indicative.

X. EXTERNAL VALIDITY: HUMAN MOBILITY TRAJECTORIES

Section VI showed that the graph term’s apparent benefit depends on the measurement-validity protocol applied before evaluation, on a single wireless deployment. This raises a natural follow-up question: is the effect specific to that deployment, or does the same linear graph state-space model (Eq. 3) generalise to a structurally different sensing domain? We apply it unchanged to human GPS mobility trajectories, substituting position (x, y) for (T, H, L) and co-presence for spatial adjacency.

A. Setup

We use GeoLife [13], GPS logs from users in and around Beijing, restricted to the 28 users carrying transportation-mode labels, a 50-day busiest-activity subset, and the Beijing metropolitan bounding box. Raw points are resampled onto a common absolute 5-second grid so different users’ trajectories are directly time-comparable, then cut into non-overlapping 100-second windows (8 observed steps, 12 held out), following the observation/prediction convention of TrajImpute [14]. Two windows sharing a global time index are a candidate scene; an edge is added when the pair’s mean position over the window is within 50 m, the radius separating genuine co-presence from same-clock-time coincidence in this data. θ is fit per transportation mode by the same pooled least-squares procedure used for the deployment (Section VI), on a 70/15/15 user-disjoint split. Missingness is simulated on the 8 observed steps under the same Easy ($m \sim \mathcal{U}\{0, \dots, 4\}$) / Hard ($m \sim \mathcal{U}\{4, \dots, 7\}$) protocols as TrajImpute. Unlike the deployment, there is no faulty subpopulation to gate against here, so this check isolates the estimator’s behaviour from the fault-detection question addressed in Sections III–V.

We report the two modes most relevant to a fleet-of-vehicles reading of the deployment result: private car and taxi. The two

TABLE XI
GRAPH-MESSAGE-PASSING GENERALISATION TO GEOLIFE TAXI AND PRIVATE-CAR TRAJECTORIES (POSITION MAE, METRES).

Mode	Protocol	No graph	Graph MP	Persist.	θ_{sp}
Car (private)	Easy	26.714	26.709	27.249	+0.0130
Car (private)	Hard	47.274	47.257	47.896	+0.0130
Taxi	Easy	35.629	35.617	36.486	+0.0626
Taxi	Hard	59.555	59.546	60.274	+0.0626

were initially merged, as the GeoLife label guide groups both under a single “driving” category; splitting them changes the private-car numbers materially, evidence the two populations follow different dynamics. Walk, bus, bike and subway were also tested and show the same qualitative pattern (graph beating persistence) but are not reported in detail here.

B. Results

Table XI shows the graph term beats persistence in both modes and both protocols, consistent with the deployment’s positive result under its own (voltage-gated) validity protocol. Taxi’s fitted $\theta_{sp} = 0.0626$ is the largest graph-coupling coefficient obtained anywhere in this study—roughly $5\times$ the deployment’s 0.0064 (Section VI) and $4.8\times$ private car’s 0.0130—consistent with commercially dispatched vehicles sharing traffic-flow correlations more tightly than private trips or a fixed indoor deployment.

This coefficient is fit from ten genuine co-presence scenes among seven training users. But the user-disjoint test split places only two taxi-labelled users in the held-out set, and they never co-occur: all 261 test scenes are singletons, so the fitted graph term has no edge to act on during evaluation. The resulting test-set MAE improvement (0.03%) is negligible despite the underlying coefficient being an order of magnitude larger than any other mode’s.

C. A second illustration of protocol-dependence

This is a second, independent demonstration of the paper’s central claim, arrived at by a different mechanism. Section VI showed the measurement-validity rule determines whether a real graph benefit is visible. Here, the composition of the evaluation split plays the analogous role: a coefficient fit correctly and substantially from training data can be invisible in a held-out test metric purely because that split happens to contain no instance of the structure the coefficient encodes. A negative or near-null result at test time does not certify the absence of the effect being measured in either case—only that the evaluation protocol failed to exercise it.

XI. DISCUSSION

Measurement-validity protocol can dominate methodological effect. The difference between the two rows of Table VII exceeds any difference we observed between estimator variants. Where a corruption mechanism is physically indexed, gating on that index is not housekeeping—it decides the result. The same hazard should be expected wherever

sensor degradation correlates with time and evaluation splits are chronological.

Detectors should be validated against instruments that do not share their information. The voltage gate appeared sound in isolation; only a voltage-free detector could establish that it misses almost nothing, and only a third arbiter could establish that its rejections were largely unwarranted. Reporting a validity rule without such a check leaves its false-rejection rate unmeasured.

Detection and reconstruction are not interchangeable. It is tempting to treat sufficiently robust reconstruction as a substitute for identifying bad data. Table IX shows otherwise on a sparse graph: corruption a deterministic detector removes exactly will defeat a median aggregator at 5% prevalence, because the aggregator’s quorum is a neighbourhood of three rather than a network of 54.

Limitations. We fit the two-parameter linear model of (3) rather than a nonlinear spatio-temporal graph network, so the 27.1% should be read as a lower bound on what the architecture can deliver. The 6 m radius was fixed a priori from connectivity and not tuned against the hold-out objective. The tightened bounds of Section V-A encode this building’s operating envelope over five late-winter weeks, not a sensor specification; the method transfers, the constants do not. Finally, the dataset’s published radio-connectivity file offers an adjacency grounded in measured link quality rather than Euclidean distance, which would be a natural comparison. The GeoLife check (Section X) is similarly bounded: 28 labelled users, 50 days, Beijing only, and as few as two users in some held-out splits; we report taxi and private car as the modes most relevant to a fleet-vehicle reading of the deployment result.

XII. CONCLUSION

We characterised battery-gated measurement validity on a widely used wireless sensor deployment and traced its consequences downstream. The corruption is a discrete rail fault rather than a drift, indexed causally by battery voltage, and a voltage-free fusion detector independently confirms the gate misses almost nothing while over-discarding 8.59% of readings. A staged cleaner exploiting this recovers 10.9% more data and improves accuracy 3.6%.

Most consequentially, the choice of validity protocol reverses the sign of a downstream methodological conclusion: relational structure in a graph Kalman filter appears 13% harmful under range filtering and $27.1 \pm 2.9\%$ beneficial under voltage gating. For practitioners, the measurement-validity rule belongs in the ablation table alongside the model.

The same conclusion holds beyond this deployment. Applying the identical model to human GPS trajectories (taxi and private-car mobility) reproduces the graph benefit, and a well-fitted coupling coefficient—the largest observed in either domain—can still be invisible in a held-out split that contains no instance of the structure it encodes: a second, independent route to the same protocol-dependence identified in this paper.

DATA AVAILABILITY

The Intel Berkeley Research Lab dataset, comprising both readings and mote coordinates, is publicly available at <https://db.csail.mit.edu/labdata/labdata.html>. All analysis code accompanies this manuscript: `prepare_intel_gss.py` (voltage-gated preprocessing), `fusion_detector.py` (Section IV), `tightened_bounds.py` (Section V-A), `staged_cleaner.py` (Section V), `gkf_experiment.py` (Section VI), `harmonic_reconstruction.py` and `lowrank_completion.py` (Sections VII–VIII), and `local_evaluation.py` (Section IX). The GeoLife check (Section X) uses `geolife_prepare.py`, `geolife_windows.py`, `geolife_split.py`, and `gkf_geolife.py`. GeoLife is publicly available from Microsoft Research; TrajImpute’s ETH/HOTEL splits and protocol code accompany [14]. The paper source and all scripts listed above are maintained at <https://github.com/viyer-research/gkf-trajectory-generalization> (repository to be made public upon publication).

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